Course Projects Guide

AI and Data Science Professional | What you build, keep and present | Updated 4 November 2026

How this course works

There are no exams, marks, grades, pass marks or weights in this course. You build practical projects and you keep them.

The check cells (OK or NOT YET), the recall questions and the exit-check questions are self-checks for you. Nobody marks them.

You finish the course by finishing your course project. Then you receive a certificate of completion from KLAi Professional Learning.

1. What you build across the course

Every lab ends with something you keep. Together they are your portfolio of practical work.

Module A: Think with data (Sessions 1 to 4)

#	Session	What you keep from the lab
1	What data science is and how to think in it	Notebook in Drive and your business question
2	Data as tables: records, variables and keys	Workbook dictionary tab: record, key, problems per tab
3	Summarising and showing data fairly	Workbook: three pivot tables and three charts
4	Uncertainty, samples and cause	Simulation tab: 200 samples of 30 orders

Module B: Python for data (Sessions 5 to 9)

#	Session	What you keep from the lab
5	Your first Python: variables, lists and error messages, next to the spreadsheet	Notebook: price list, one error read and fixed
6	Decisions, repeats and functions: if, for, and a tested function	Notebook: a tested loyalty_tier function
7	pandas I: load, look and filter a table of 30,229 orders	orders_profile.csv: a data profile
8	pandas II: cleaning the mess, with a log and a reconciled total	orders_clean.csv, cleaning log, reconciliation note
9	pandas III: group, join and reshape	Revenue tables and a join log with row counts

Module C: Analyse and explain (Sessions 10 to 13)

#	Session	What you keep from the lab
10	SQL for analysts: ask a database the questions you ask pandas	Query book: ten SQL queries with pandas proofs
11	Exploring and visualising data with Python: from a new file to a finding	Notebook: eight charts and three findings
12	Inference: is the difference real, or could it be luck?	Notebook: the offer test five ways, a decision note
13	Regression: explaining and predicting a number	regression_summary.csv and a one-page report

Module D: Machine learning (Sessions 14 to 19)

#	Session	What you keep from the lab
14	Machine learning: the workflow, from loans data to a first model	loans_prepared.csv and a three-sentence note

#	Session	What you keep from the lab
15	Classification: predicting a yes or no answer with logistic regression and a decision tree	loan_risk_scores.csv and a threshold note in UGX
16	Judging a model: metrics, leakage and imbalance	model_comparison.csv and a note on the leaked column
17	Forests, boosting and overfitting: a stronger model without fooling ourselves	Model comparison table and a decision memo
18	Finding structure without labels: customer segments and unusual orders	customer_segments.csv and unusual orders to review
19	Forecasting and a first end-to-end model: baseline, time-based split and Pipeline	Baseline, tested forecast, Pipeline, manager note

Module E: Neural networks and text (Sessions 20 to 22)

#	Session	What you keep from the lab
20	Neural networks and deep learning, without the maths wall	Notebook, results table and a recommendation
21	Text as data and meaning as numbers	feedback_scored.csv and a three-sentence finding
22	Language models and building with them	eval_results.csv and audit_log.csv

Module F: Ship it and present (Sessions 23 to 24)

#	Session	What you keep from the lab
23	Ship it: reproducible work, a small app and responsible use	Shipping folder: app.py, saved model, model card
24	Present your course project and plan your path from here	Presentation sheet and a six month learning path

2. Three project milestones

A project milestone is a practical piece of work that you produce in a session and show to your tutor. The tutor reviews it with you and gives feedback: what works and what to improve. A milestone is never scored. It is a step toward your course project.

Project milestone 1, Session 8: clean a raw file and reconcile it

A raw order export is cleaned one defect at a time. You produce four things.

You produce	A finished version has
The cleaned notebook	Every check cell says OK and the run ends with All checks pass.
The data cleaning log	It has one row for each of the seven steps, and the last row ends at 41,167.
A reconciliation note	One or two sentences name the reconciliation difference (132,690 UGX) and its likely cause.
The files orders_clean.csv and cleaning_log.csv	They are saved from df and the log, and open in Excel or Sheets.

A reconciliation difference you can explain is worth more than one you have hidden. The explanation in your note is in your own words.

Project milestone 2, Session 13: a one-page analysis report

One page. A manager who has never seen the data should understand it. You also write your project question in this session (section 3).

Section	What you write
Question	One sentence naming the target, the predictors and the decision.

Section	What you write
Model	The multiple regression and the unit of every predictor.
Findings	Three coefficients as sentences, with intervals, and R squared.
Check	What the residual plot showed and what you would do about it.
Limits	What the model cannot prove, and the test RMSE against the baseline.

Project milestone 3, Session 19: a tested model

You forecast clinic cases by district and month, and keep your notebook and the results file it writes. It shows:

- a baseline (a seasonal naive forecast) and a model scored on months it never saw, split by date at 2025;
- the error in the units of the business (cases), reported with MAE, RMSE and MAPE;
- a scikit-learn Pipeline that packs the modelling job into one object;
- three sentences for a district manager: whether the model beats the baseline, by how much, and what could mislead.

The same three moves apply to your course project: a baseline, a score on records the model never saw, and an error in business units. Check for data leakage.

3. The course project

You choose a question and a dataset and carry them through the second half of the course. A good question names the target, the predictors, the data and the decision. Session 13 gives examples.

Session	Course project step
13	Write and scope the question (Project milestone 2)
14	Choose the dataset
19	Baseline and tested model (Project milestone 3)
23	Learn to publish the app and model card; apply it to your course project before Session 24
24	Present for ten minutes

Dataset options

- Lakeside Mart: orders, order lines, branches, customers and weekly branch figures (a fictional retailer).
- Nile Microfinance loans: 8,000 loans, and whether each defaulted (fictional).
- District Clinics: monthly clinic visits by district (fictional).
- Customer feedback messages: 800 short messages (fictional).
- Your own question about one of the tables above. If you want to use data from your workplace, ask your employer in writing, remove names and phone numbers first, and never put it in a public repository.

What your course project contains

1. A one-sentence question and the decision it serves.
2. The data cleaned, with a data cleaning log.
3. One analysis that answers part of the question.
4. One model with a baseline model and a metric chosen for the cost of a mistake, scored on records it has not seen.
5. The limits: what the data cannot show, who could be harmed by an error, and which groups you compared.
6. A small app (Session 23) with a model card and a data protection checklist.
7. A ten-minute presentation (Session 24).

4. How to prepare and present it

Five slides in ten minutes: question (1 min), data (2 min), evidence (3 min), risk (2 min), decision (2 min). Evidence gets the most time.

- Tell the result as a story in three parts: the answer, the evidence beside a baseline model, and the limits.

- Write an executive summary of four sentences: the answer, the evidence, the main limit and the decision you ask for. It becomes your opening.
- Say the limits before the result. People trust a speaker who names them first.
- Use the real term once, then say what it means for a decision. Recall, for example, is how many of every 100 defaults you flag.
- Rehearse aloud with a timer. Most people run long.
- Prepare three answers: one on method, one on the metric and one on risk. An answer that says a check is not finished is a good answer.

Session 24 is a planning and rehearsal lab in a Google Sheet. If you have no project of your own, you present the Nile Microfinance loan story from the slides.

5. The project checklist: feedback for yourself and for a partner

Some people call a list like this a rubric. Here it is a checklist of four qualities that a finished project has. Use it on your own project before you present. In the Session 24 lab your partner uses the same four rows to give you feedback: Yes or Not yet for each row, and one comment that points to a named slide. Nothing is scored and nothing is weighted.

Quality	A finished project has	A common gap to fix
Question and evidence	A one-sentence question. Every claim points to data.	No clear question, or claims without data.
Metric and baseline	A metric chosen for the cost, on unseen data, beside a baseline model.	One accuracy number and no baseline.
Limits and fairness	Limits first. Risks named. Groups compared. Data duties checked.	Limits missing, or said late.
Clarity	Five slides, ten minutes, plain words, one decision.	Over time, jargon, no decision.

When you receive feedback, write it down and do not defend yourself. Ask a question if you need to understand it.

6. What the check cells mean

Every Python notebook has check cells. After each answer you run a line such as check("label", value, expected). It prints OK or NOT YET, and the last cell, show_checks(), prints "All N checks pass" or "k of N checks pass". They are self-checks. Nobody marks them.

What OK means	Your value is within a small tolerance of the expected value (0.01 unless the lab says otherwise). Your cell ran and produced that number.
What NOT YET means	Your value is outside the tolerance, or the cell has not been filled in, or it raised an error. It means look again.
What a check cannot tell you	That you understand the method or that your reasoning is right. A correct number can come from a wrong method, and a wrong number can come from a harmless rounding or a setup problem.
Only the checks you ran count	"All 7 checks pass" means the 7 checks you ran passed. A check you skipped is not counted. Run the cells in order and look at the total.
Libraries differ a little	Models from scikit-learn can differ from the slides by a few records or a unit in the second decimal. The checks use a tolerance for this. Random exercises (Session 4) use ranges.
A learner notebook starts with NOT YET	That is expected. The answer cells hold answer = None until you fill them in. The solution notebook ends with "All N checks pass".
Sheets sessions	Sessions 1 to 4 and 24 have a "Check your numbers" table on the slides. You compare your sheet with it by eye. There are no check cells.

7. The certificate of completion

You receive a certificate of completion from KLAi Professional Learning when you complete your course project. There are no criteria beyond that: no marks, no pass mark and no weights. It says that you finished the course and your project. It is not an external credential or a university accreditation.

8. Working rules

- Use the course data. It is fictional and generated.
- A free chat assistant may explain an error or suggest code. Keep a short audit log: what you asked, what it gave, how you checked it, and whether you kept it. You must be able to explain every line in your project.
- List what you did not check. A stated limit builds trust. A hidden one costs it.
- Your published work is public. Check it for personal data, passwords and keys before you push it to GitHub.
- Never paste real customer or patient data into a notebook, an assistant or an app you share.